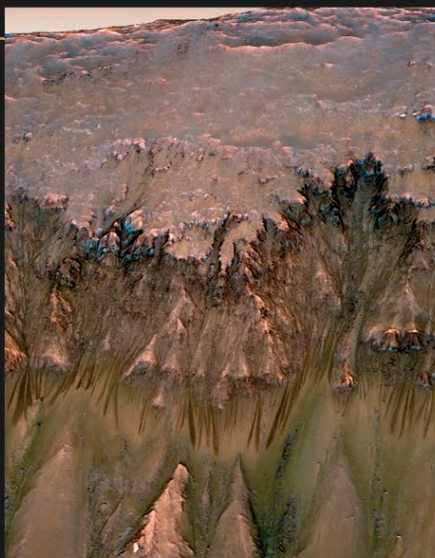




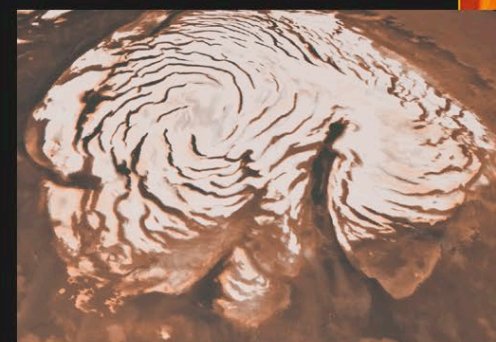
WATER ON MARS

Scientists have been hoping to establish whether water is present on Mars, since this is essential for the development of life. Liquid water is not present, as this is a cold planet, where water can exist only as ice or vapor. The latter can form low-lying mists and fogs and freezes into a thin layer of white water ice on the rocks and soil when the temperature falls. However, dry river beds, valleys, and ancient floodplains bear witness to the presence of large amounts of fast-flowing water on the surface 3–4 billion years ago, when Mars was a warmer, wetter world

THE CASE FOR FLOWING WATER

Images from Mars Global Surveyor have been processed to create this 3-D view of seasonally variable features on the inner slopes of the crater Newton, possibly created by running water.

with a thicker atmosphere. Some of that water remains today in the form of ice, which is present both underground and in the polar ice caps. The ice caps wax and wane with the Martian seasons and are composed of varying amounts of water ice and frozen carbon dioxide.



NORTH POLAR ICE CAP

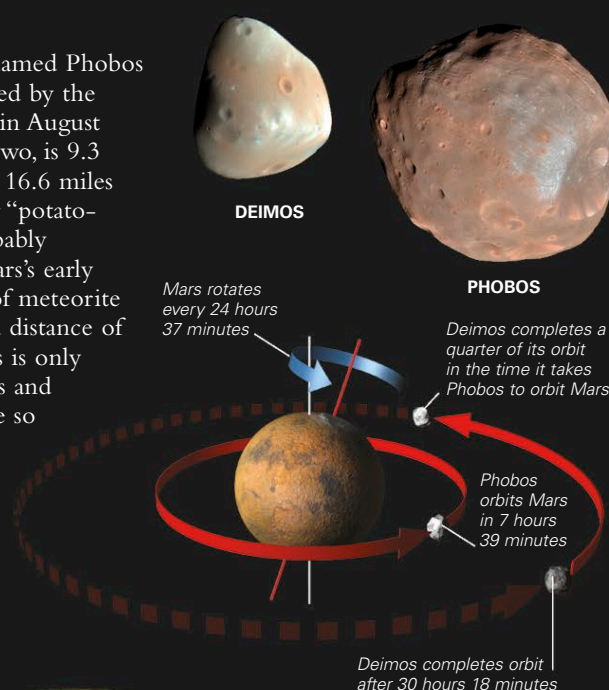
The north polar cap of Mars, seen by Mars Global Surveyor, is about 600 miles (1,000 km) across and is cut by spiral-shaped troughs. At center right is Chasma Boreale, a valley about the length of Earth's Grand Canyon.

MOONS

Mars has two small, dark moons named Phobos and Deimos, which were discovered by the American astronomer Asaph Hall in August 1877. Deimos, the smaller of the two, is 9.3 miles (15 km) long and Phobos is 16.6 miles (26.8 km) long. Both are irregular “potato-shaped” rocky bodies and are probably asteroids that were captured in Mars’s early history. They both bear the scars of meteorite battering. Deimos orbits Mars at a distance of 14,580 miles (23,500 km). Phobos is only 5,830 miles (9,380 km) from Mars and getting closer; eventually, it will be so close that it will either be torn apart by Mars’s gravity field or collide with the planet.

MOONS’ ORBIT

Phobos and Deimos both follow near-circular orbits around Mars, and both exhibit synchronous rotation. From Mars, Phobos rises and sets three times every Martian day.



GEOGRAPHY

The first reliable maps of Mars were made in the late 19th century, when astronomers drew what they observed through their telescopes. Today’s maps are based on data collected by space probes such as Mars Global Surveyor, which obtained 100,000 photos of Mars and completed a survey of the planet, and Mars Express, which is imaging the entire surface. The following terminology is used for the surface features: lowland plains are termed planitia; high plains, planum; extensive landmasses, terra; and mountains or volcanoes, mons. A chasma is a deep, elongated, steep-sided depression, and a labyrinthus is a system of intersecting valleys or canyons. Individual names are allocated depending on the type of feature. Large valleys (vallis) are named after Mars in various languages and small ones after rivers. Large craters are named after past scientists, writers, and others who have studied Mars; smaller craters are named after villages. Other features are named after the nearest albedo feature on the early maps.

